WORKSHEET 6.10

Rational Integrands: Division and Completing the Square

Unit 6: Integration and Accumulation of Change

Key ideas

- If the denominator is a single term, **divide each term** of the numerator by it.
- If the numerator's degree is at least the denominator's, do **long division** first.
- $\displaystyle\int \frac{1}{x^2+a^2}\,dx = \frac{1}{a}\arctan\left(\frac{x}{a}\right)+C.$
- A quadratic denominator with no real zeros: **complete the square**, then use the arctangent form.

Worked examples

Example 1 · Dividing term by term

Find $\displaystyle\int \frac{x^3-2x+4}{x^2}\,dx.$

Solution. $\displaystyle\int \left(x-\frac{2}{x}+4x^{-2}\right)dx.$

$\displaystyle\frac{x^2}{2}-2\ln|x|-\frac{4}{x}+C.$

Example 2 · Long division

Find $\displaystyle\int \frac{x^2+1}{x-1}\,dx.$

Solution. $\displaystyle\frac{x^2+1}{x-1} = x+1+\frac{2}{x-1}.$

$\displaystyle\frac{x^2}{2}+x+2\ln|x-1|+C.$

Example 3 · Completing the square

Find $\displaystyle\int \frac{1}{x^2+4x+5}\,dx.$

Solution. $x^2+4x+5=(x+2)^2+1.$

$\arctan(x+2)+C.$

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Practice

CORE

1. Find $\int \frac{x+3}{x}\,dx$.

2. Find $\int \frac{1}{x^2+25}\,dx$.

3. Find $\int \frac{5}{x+2}\,dx$.

4. Complete the square: $x^2-6x+13$.

EXAM LEVEL

5. Find $\int \frac{x^2+3x}{x+2}\,dx$.

6. Find $\int \frac{1}{x^2-6x+13}\,dx$.

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7. Find $\int \frac{2x+3}{x^2+1}\,dx$.

8. Evaluate $\int_0^{\sqrt{3}} \frac{1}{1+x^2}\,dx$.

Question 9 · Free-response practice

Let $q(x) = x^2 + 4x + 8$.

(a) Find $\int \frac{4x+8}{q(x)}\,dx$.

(b) Find $\int \frac{1}{q(x)}\,dx$.

(c) Use parts (a) and (b) to find $\int \frac{4x+9}{q(x)}\,dx$.

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Worked solutions

1. $x + 3\ln|x| + C$.
2. $\dfrac{1}{5}\arctan\left(\dfrac{x}{5}\right) + C$.
3. $5\ln|x+2| + C$.
4. $(x-3)^2 + 4$.
5. $\dfrac{x^2+3x}{x+2} = x + 1 - \dfrac{2}{x+2}$: $\dfrac{x^2}{2} + x - 2\ln|x+2| + C$.
6. $(x-3)^2 + 4$: $\dfrac{1}{2}\arctan\left(\dfrac{x-3}{2}\right) + C$.
7. $\displaystyle\int \frac{2x}{x^2+1}\,dx + 3\int \frac{1}{x^2+1}\,dx = \ln\left(x^2+1\right) + 3\arctan x + C$.
8. $\Big[\arctan x\Big]_0^{\sqrt{3}} = \dfrac{\pi}{3}$.
9. (a) The numerator is $2q'(x)$: $2\ln\left(x^2+4x+8\right) + C$.

   (b) $q(x) = (x+2)^2 + 4$: $\dfrac{1}{2}\arctan\left(\dfrac{x+2}{2}\right) + C$.

   (c) $4x + 9 = (4x+8) + 1$: $2\ln\left(x^2+4x+8\right) + \dfrac{1}{2}\arctan\left(\dfrac{x+2}{2}\right) + C$.

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